



SYLLABUS
COLLEGE OF ARTS AND SCIENCES
DEPARTMENT OF COMPUTER SCIENCE
CSC 4520/6520: DESIGN & ANALYSIS OF ALGORITHMS
ACADEMIC TERM: SPRING 2026

INSTRUCTOR INFORMATION

Instructor:

Abdullah Bal, PhD

Office: 25 Park Place, Office 739

E-mail: abal@gsu.edu

Office hours: Tuesday-Thursday 1.30 pm-2.30 pm or by appointment

Teaching assistant and grader:

Ajay Varada

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COURSE INFORMATION

Course hours: MW 2:30 pm – 4.15 pm

Course location: Classroom South 427

This course will teach you how to analyze problems and solve them using computational procedures called algorithms. We will cover many algorithms foundational to Computer Science and learn their uses, advantages, and disadvantages. Most importantly, we will develop skills and intuition to help you design efficient algorithms for new problems.

Course Outcomes:

By the end of the course, you'll be able to

- Use brute force, decrease-and-conquer, divide-and-conquer, transform-and-conquer, greedy programming, and dynamic programming techniques to solve problems.
- Implement and employ a variety of advanced data structures to solve problems, including graphs, self-balancing trees, and trees.
- Implement a variety of algorithms to sort, search, and traverse data.
- Determine the time and space efficiency of algorithms using Big O, Big Theta, and Big Omega

notation in terms of common complexity classes.

- Identify the appropriate use cases of these algorithms, data structures, and programming techniques and evaluate the tradeoffs of possible solutions, including time and space complexity, to decide on an approach to a problem.

The content and skills in this course have a wide variety of applications in both academia and industry. Pragmatically, many tech companies expect interview candidates to be familiar with these topics and skills. At the same time, this course provides a mental framework that lets you solve hard problems—the problem-solving techniques taught here show up in various subfields in Math, Engineering, and Computer Science.

Prerequisites and Expected Background: You should have some knowledge in most of the following areas (not limited to):

CSC 2720 Data Structures – You should be familiar with coding in Python or Java, data structures such as Linked Lists, Arrays, Trees, and Hash Tables, programming techniques like recursion, as well as finding the runtime of a function via Big-O. We will review essential concepts as we go, so don't worry if you haven't mastered everything yet.

Math 3020/3030 – You should feel comfortable with function notation (e.g. $f(x) = x^2$, logarithmic functions (e.g. $f(x) = \log_2 x$), and graphs of mathematical functions. Beyond that, we'll cover the necessary mathematical intuition as we go.

Textbook

The textbook is optional. While it may be useful for perspective and context into most of our topics, it will not cover the exact same set of topics we'll traverse.

- *Introduction to The Design and Analysis of Algorithms (3rd Edition)* by Anany Levitin, Pearson.
- *Introduction to Algorithms*, by T.Cormen, C. Leiserson, R.Rivest and C.Stein, MIT Press, 3rd Edition.

Recommended:

- *Algorithms* by R. Sedgewick, K. Wayne
- *The Algorithm Design Manual* by S. Skiena
- *An Introduction to Bioinformatics Algorithms* by N. Jones, P. Pevzner

COURSE SCHEDULE

The chapter numbers below are from Levitin's book.

Week	Topic	Activities
1 Jan 12-18	Introduction to Design and Analysis of Algorithms (ch1)	
2 Jan 19-25	Complexity Analysis (ch2)	No class on Monday, January 19
3 Jan 26- Feb 1	Brute Force and Exhaustive Search (ch3)	
4 Feb 2-8	Graph Traversal (ch3)	HW-1
5 Feb 9-15	Decrease-and-Conquer (ch4)	Quiz 1, February 11 Wed.
6 Feb 16-22	Decrease-and-Conquer (ch4)	
7 Feb 23- Mar 1	Midterm Exam 1	February 25, Wednesday
8 Mar 2-8	Divide-and-Conquer (ch5)	HW-2
9 Mar 9-15	Transform-and-Conquer (ch6)	Quiz 2, March 11 Wed.
10 Mar 16-22	Spring Break	
11 Mar 23-29	Transform-and-Conquer (ch6)	
12 Mar 30- Apr 5	Midterm Exam 2	April 1, Wednesday
13 Apr 6-12	Space-time trade-offs (ch7)	HW-3
14 Apr 13-19	Dynamic Programming (ch8)	Quiz 3, April 15 Wed.
15 Apr 20-26	Greedy algorithms (ch9)	
Apr 29, Wednesday	Final Exam	Exam Time: 13.30-16.00

GRADING

1. Homework Assignments

Assignments permit you to practice and apply what we are learning in class. All homework should be uploaded on iCOLLEGE drop box into the corresponding folders. The system also manages cut-off time, so 1 minute late is still late! Prepare accordingly.

Assignments should be TYPED, NOT HANDWRITTEN and/or SCANNED;

Submissions should be either MS Word/OpenOffice or pdf documents, as other formats might not be supported for reading.

2. Tests

All tests will be given in class and held under closed-note conditions.

3. Extra credit

There are no extra credit assignments in this course.

4. Course Grade

At the end of the course, your weighted numerical average will be converted into letter grades based on the following scale. If you are in the range below you are guaranteed that letter grade, but your grade might be improved with a +/- depending on the distribution. For example, a 79% is guaranteed a C, but could be upgraded to a C+ or a B-.

90 -100% = A

80-89%. = B

70 -79% = C

60 - 69 = D

0 - 59% = F

Your final course grade will be computed as follows:

Homework (3) 10%

(For graduate students the last homework will be a project)

Quiz (3) 30%

Tests (2) 30%

Final exam..... 30%

The scheme could be changed at the instructor's discretion, and you will be notified in advance.

The TA is responsible for grading all assignments, quizzes and exams. While the instructor might provide the grader with guidelines (for example, that a question is worth 10 points), it is the TA's job to determine correctness of your answer. The TA is authorized to adjust grades if there has been a mistake in the grading (for example, if the grades were added incorrectly). Therefore, **if you have questions regarding how your work was graded** (such as "why were 3 points taken off for this problem?"), **please contact the TA.** Any questions regarding graded work must be brought to the TA's attention within 2 weeks of the date on which the work was first returned to the class (or grades were reported to you by e-mail).

POLICIES

1. Class Participation

Full participation is expected. Be ready to answer questions, work with others in group, solve problems on board in class under instructor's guidance, and other activities.

2. Late Assignments & Missed Exams

In the event of a late assignment submission, there remains a chance to attain a grade of 80% by submitting the assignment within a one-day grace period. However, no credit will be given for work submitted after the one-day deadline.

In addition, makeup tests are not given. Exceptions to this policy will be made only in case of a family or medical emergency, which must be clearly documented.

3. Attendance

Class attendance is required. Also, if you are absent, it is your responsibility to find out what you missed and to come to the next class prepared.

4. Academic Integrity

The course is designed so that you do well when you put in the effort. The instructors will make every effort to help you succeed. Should you find yourself in a position where you feel trapped or

encouraged to cheat, we hope you will email us instead to ask for help. That being said, here are the expectations around academic honesty.

Cheating

For homework, copying solutions from any source (students, websites, textbooks), whether wholesale, in part, or modifying the code afterwards, is strictly considered cheating. **Providing solutions to other students or third-party websites, in part or in whole, is also considered cheating.**

Exams are independent exercises. Receiving help in any form outside of asking questions of the instructors and consulting your own written notes/the course materials is strictly prohibited.

Collaboration

Note that the above policy does not mean that you have to work alone the entire semester. For homework problems, you're allowed to discuss approaches, provide hints, and key insights with up to five classmates, but must write up solutions on your own. Please cite any students, web sources, etc. that you consulted while solving problems. **You should never possess solutions to exact homework questions other than those solutions you have written yourself.**

5. Closing of the University

Sometimes GSU closes because of weather conditions. To find out if the university is closed, go to **www.gsu.edu** or listen for an announcement on local TV or radio channels. If class is canceled, planned tests will take place during the next class session.

6. Cell Phone Policy

Please turn off your cell phones or put them on silent operation before entering the class. Make sure to take good notes to succeed in this class.

7. Students with Special Needs

Students who need special accommodations should arrange a meeting with the instructor as soon as possible. Bring a copy of your Student Accommodation Form to the meeting. If you do not have this form, make an appointment with the university's Office of Disability Services, 230 New Student Center, 404-413-1560.

This syllabus represents a general plan for the course, deviations may be necessary.